Method of Differentiation

Mind map

First principle
- $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$
- Dot $= \frac{d}{dt}$

Standard derivatives
- 19 results
- Co-functions: minus sign

Product and quotient
- $(fg)' = f'g + fg'$
- $\left(\frac{f}{g}\right)' = \frac{gf'-fg'}{g^2}$
- Simplify first

Chain rule
- $f'(g(x))\,g'(x)$
- $\frac{dy}{dx} = 1 \Big/ \frac{dx}{dy}$

Log differentiation
- Products, f^g
- $f^g[g' \ln f + \frac{gf'}{f}]$

Parametric
- $\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}}$
- $y'' = \frac{\dot{x}\ddot{y}-\dot{y}\ddot{x}}{\dot{x}^3}$

f w.r.t. g
- $\frac{f'(x)}{g'(x)}$
- ITF: substitute first

Implicit
- $-\frac{f_x}{f_y}$
- Homogeneous: $y' = \frac{y}{x}$

Inverse function
- $g'(b) = \frac{1}{f'(a)}$
- $f(a) = b$ by inspection

Higher order
- $y_2 = \frac{d}{dx} y_1$
- Prove $Py_2 + Qy_1 + Ry = 0$
- $\frac{d^2x}{dy^2} = -\frac{y_2}{y_1^3}$

Analysis of ITF
- Piecewise by $x = \tan\theta$
- Limits, f', f'', graph

Miscellaneous
- Differentiate identities only
- $a_r = \frac{f^{(r)}(0)}{r!}$
- $(x-a)^2 \mid f$: $f(a) = f'(a) = 0$

Determinants
- One row at a time
- Add the determinants

L'Hospital rule
- $\frac{0}{0}, \frac{\infty}{\infty}$: $\frac{f'}{g'}$
- Repeat if needed
- $0^0, \infty^0$: take ln

Integrals
- $\frac{d}{dx}\int f = f$
- Leibniz: $f(h)h' - f(g)g'$